

Tao Huang

Ph.D. Candidate in Computer Science and Technology

School of Computer Science & Technology, Beijing Jiaotong University (BJTU), Beijing, China

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EDUCATION

Ph.D. Candidate in Computer Science and Technology

Sept. 2023 – June 2028

Beijing Jiaotong University (BJTU), Beijing, China

– Advisor: Prof. Liping Jing; Co-advised by: Dr. Rui Wang

– Research Focus: Trustworthy AI: Uncertainty Learning and Adversarial Learning

Joint Ph.D. Student in Computer Science

Oct. 2027 – Sept. 2028

National University of Singapore (NUS), Singapore

– Advisor: Prof. Jin Song Dong

– Research Focus: Trustworthy AI: Formal Verification

B.S. in Mathematics and Applied Mathematics

Sept. 2019 – June 2023

Beijing Jiaotong University (BJTU), Beijing, China

– GPA: 3.77/4.0 (Rank: 4/15)

PUBLICATIONS

Detecting Misbehaviors of Large Vision-Language Models by Evidential Uncertainty Quantification

Tao Huang, Rui Wang, Xiaofei Liu, Yi Qin, Li Duan, Liping Jing

The 14th International Conference on Learning Representations (ICLR 2026, CCF A)

Accepted

Learning Universal Semantic Simplex for Jailbreaking Large Vision-Language Models

Tao Huang, Rui Wang, Xiaofei Liu, Xiaomeng Li, Yi Qin, Tong Xiao, Liping Jing

ACM Multimedia (ACM MM 2026, CCF A)

Accepted

Visual Hallucination Detection in Large Vision-Language Models via Evidential Conflict

Tao Huang, Zhekun Liu, Rui Wang, Yang Zhang, Liping Jing

International Journal of Approximate Reasoning (IJAR 2025, JCR Q2, CCF B)

Accepted

Accompanying dataset PRE-HAL released on Hugging Face (19,000+ downloads)

Learning Robust Adversarial Simplex for Black-Box Attack

Tao Huang, Zhengwei Fang, Rui Wang, Yi Li, Lei Shi, Yi Qin, Tong Xiao, Xiaofei Liu, Huafeng Liu, Michael K. Ng, Liping Jing

International Journal of Computer Vision (IJCV, JCR Q1, CCF A)

Under Review

Transferable Targeted Adversarial Attacks via Implicit Ensemble

Tao Huang, Yi Li, Rui Wang, Zhengwei Fang, Weihao Cao, Yi Qin, Tong Xiao, and Liping Jing

IEEE Transactions on Neural Networks and Learning Systems (TNNLS, JCR Q1, CCF B)

Under Review

SAM Encoder Breach by Adversarial Simplicial Complex Triggers Downstream Model Failures

Yi Qin, Rui Wang, Tao Huang, Tong Xiao, Liping Jing

International Conference on Computer Vision (ICCV 2025, CCF A)

Accepted

Strong Transferable Adversarial Attacks via Ensembled Asymptotically Normal Distribution Learning

RESEARCH EXPERIENCE

Uncertainty Quantification & Learning Based on Evidence Theory

Sep. 2024 – Present

Supervisors: Prof. Liping Jing & Dr. Rui Wang

- Proposed a post-hoc uncertainty quantification framework based on Dempster-Shafer Theory, enabling rigorous distinction between model conflict and data ignorance for detecting hallucinations and jailbreaking in large vision-language models.
- Designed an evidence-based training mechanism with a rectified loss function to endow neural networks with inherent uncertainty awareness during feature learning.
- Developed and publicly released the PRE-HAL dataset on Hugging Face, downloaded over 18,000 times.

Adversarial Transferability Based on Optimization & Flatness

Sep. 2022 – Present

Supervisors: Prof. Liping Jing & Dr. Rui Wang

- Formulated adversarial attacks as constrained optimization problems, leveraging flat optima and simplex optimization to enhance attack transferability across models.
- Developed a universal perturbation strategy (LUP) for jailbreaking large vision-language models using simplex optimization, outperforming complex multi-modal perturbation methods.

Mathematical Modeling Contest (MCM/ICM)

Feb. 2022

Team Leader & Main Modeler

- Developed evaluation models using PCA and AHP, and established a bi-objective optimization model solved via gradient descent, achieving the Finalist (Top 0.3%) award.

HONORS & AWARDS

Finalist (Top 0.3%), MCM/ICM (Mathematical Contest in Modeling)	2022
Research Innovation Scholarship, Beijing Jiaotong University	2023
Second Class Academic Scholarship, Beijing Jiaotong University (Ph.D.)	2023, 2024

ACADEMIC SERVICE

Conference and Journal Reviewer, ACM MM 2025-2026, CVPR 2026, ECCV 2026, CVIU, Pattern Recognition	2025–2026
External Reviewer, CVPR 2024, ICCV 2025	2024–2025
Teaching Assistant, Machine Learning (Prof. Liping Jing), BJTU	Fall 2023
Laboratory Service, Server Administration and Maintenance, BJTU	2023 – Present